Cheat Sheet: Plotting with Matplotlib

Plotting with Matplotlib using Pandas

Line Plot

Shows trends and changes over time

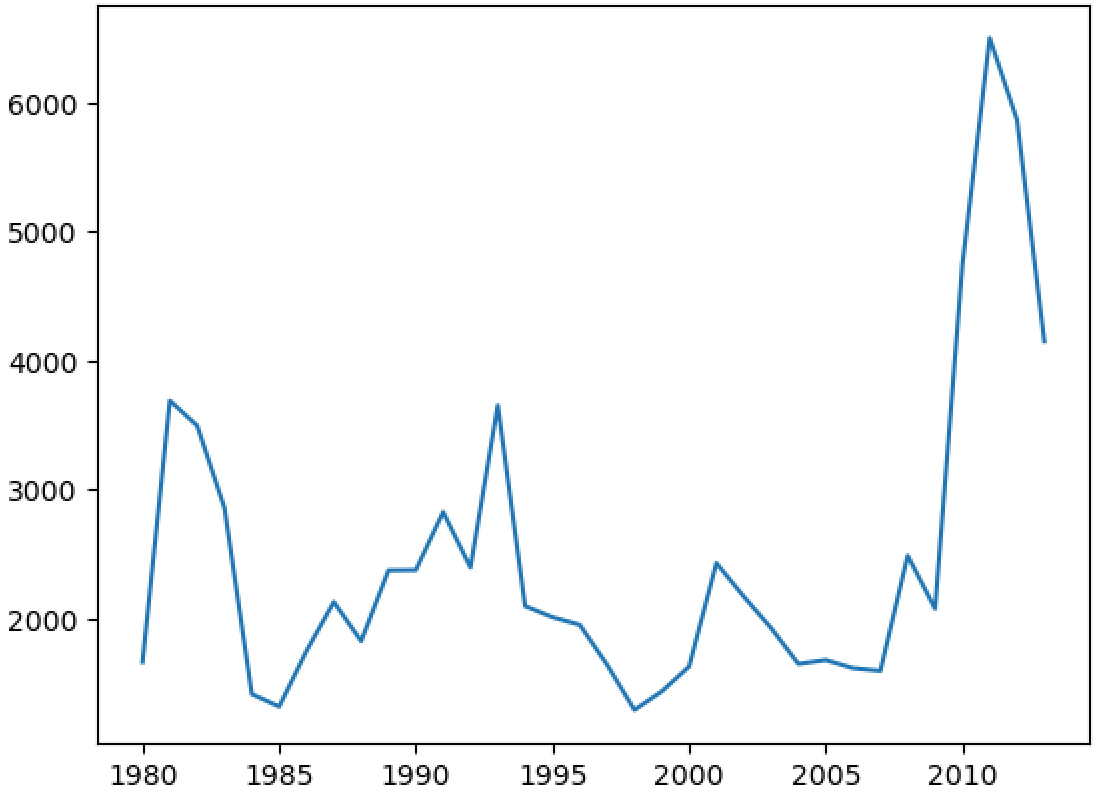
Pandas Function:

`DataFrame.plot.line()DataFrame.plot(kind = ‘line’)`

Example:

`df.plot(x=’year’, y=’sales’, kind=’line’)`

Visual:



Area Plot

Displays data series as filled areas, showing the relationship between them

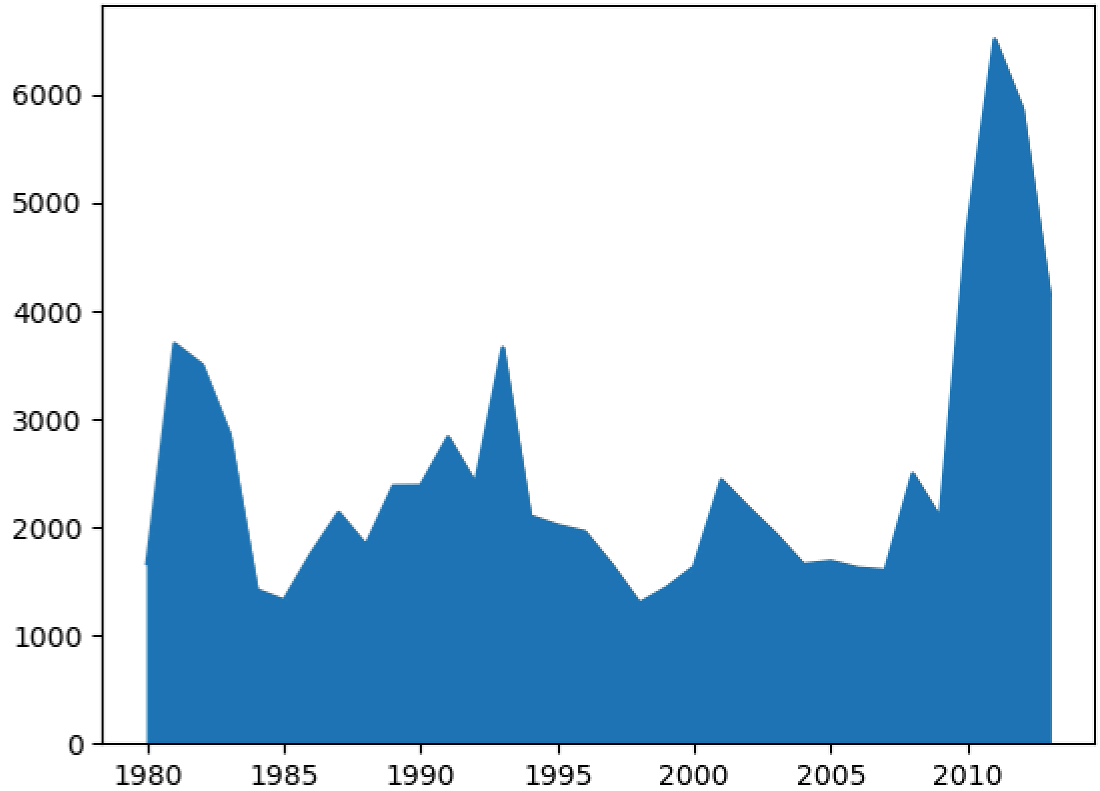
Pandas Function:

```
DataFrame.plot.area()
DataFrame.plot(kind = 'area')
```

Example:

```
df.plot(kind='area')
```

Visual:



Histogram

Displays bars representing the data count in each interval/bin

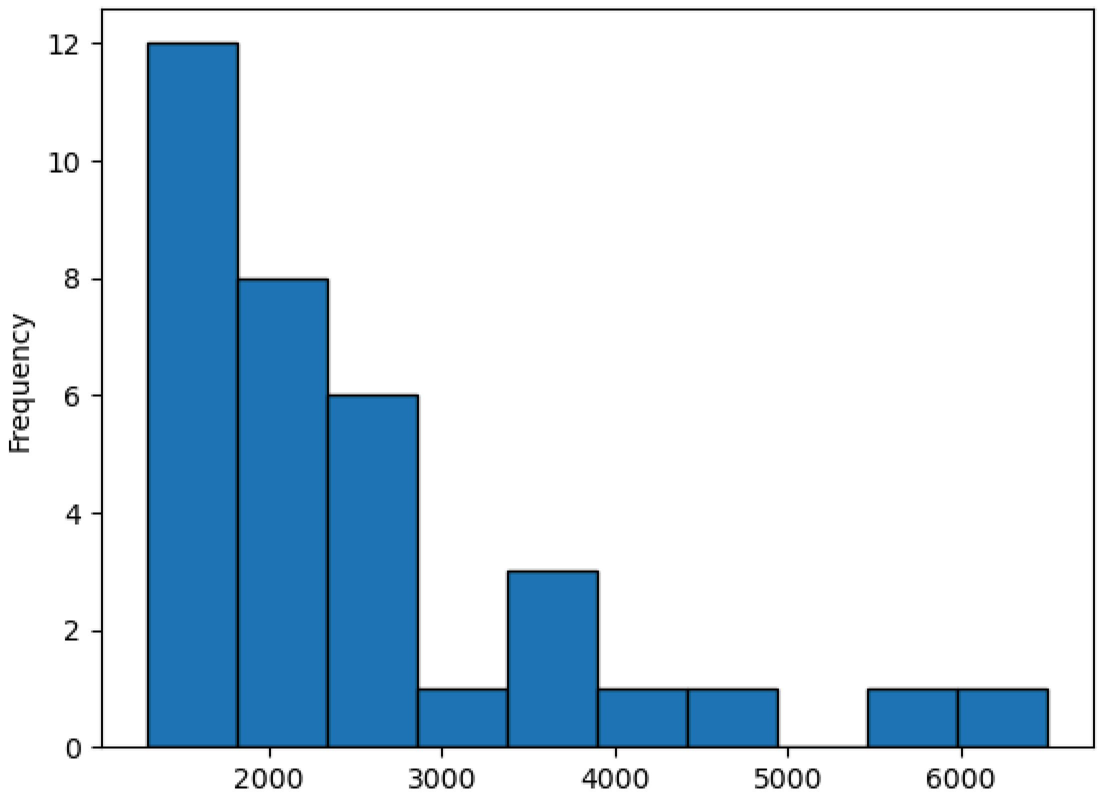
Pandas Function:

```
Series.plot.hist()
Series.plot(kind = 'hist', bins = n)
```

Example:

```
s.plot(kind='hist', bins=10)
df['age'].plot(kind='hist', bins=10)
```

Visual:



Bar Chart

Displays data using rectangular bars

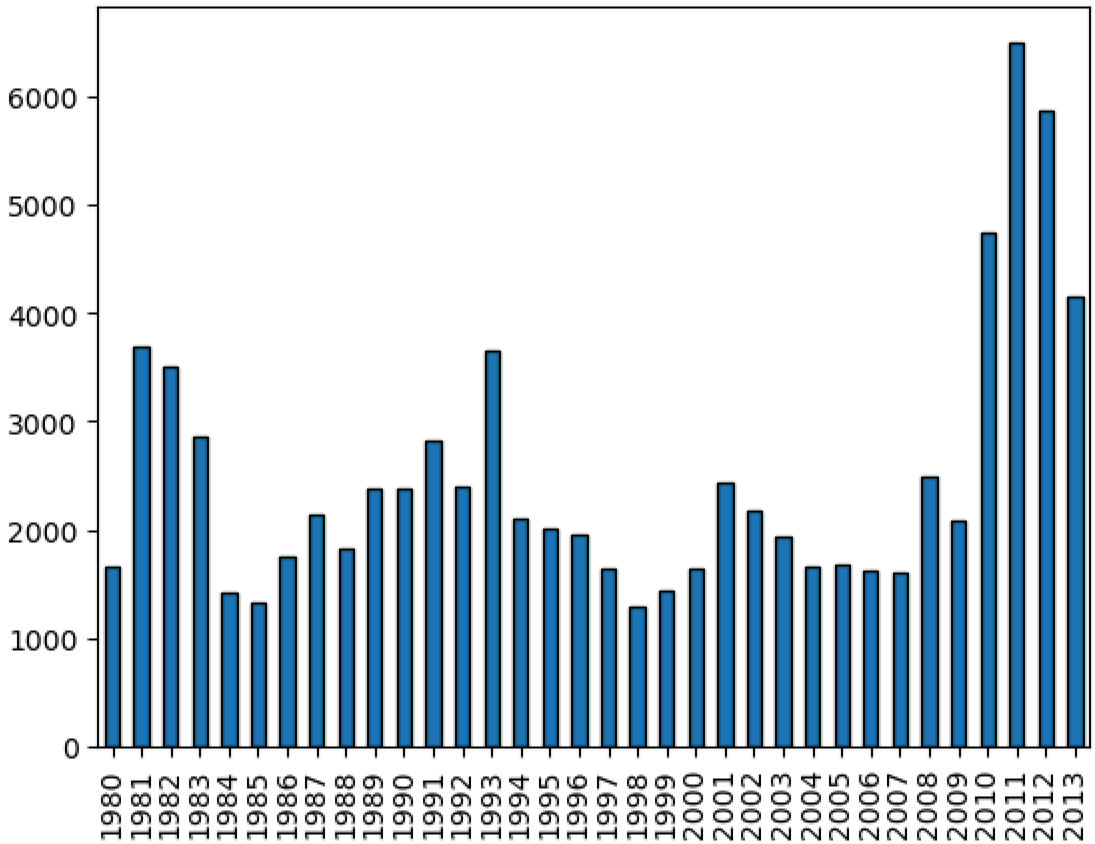
Pandas Function:

```
DataFrame.plot.bar()
DataFrame.plot(kind = 'bar')
```

Example:

```
df.plot(kind='bar')
```

Visual:



Pie Chart

Displays data as a circular plot divided into slices, representing proportions or percentages of a whole

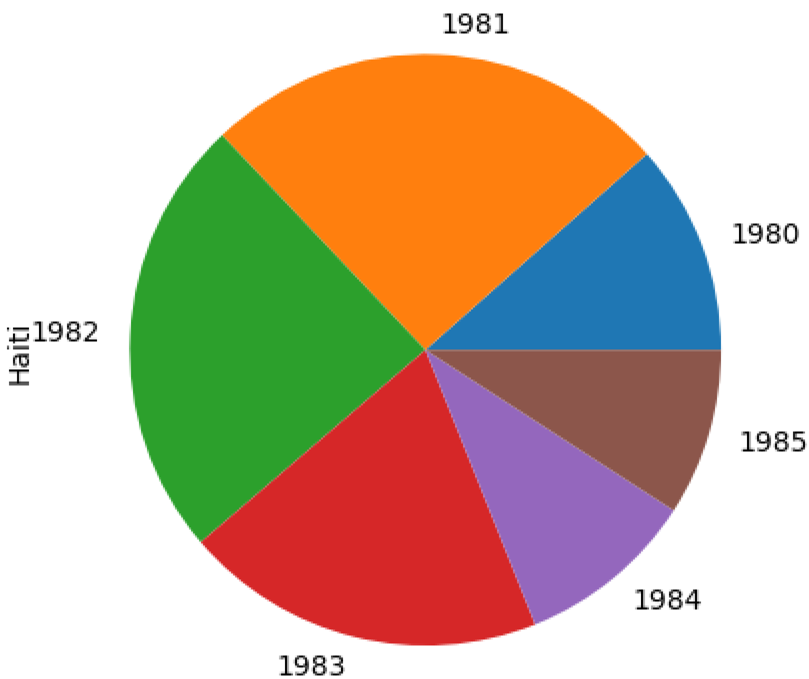
Pandas Function:

```
Series.plot.pie()
Series.plot(kind = 'pie')
DataFrame.plot.pie(y, labels)
DataFrame.plot(kind = 'pie')
```

Example:

```
s.plot(kind='pie', autopct='%1.1f%%')
df.plot(x='Category', y='Percentage', kind='pie')
```

Visual:



Box Plot

Displays the distribution of a dataset along with key statistical measures

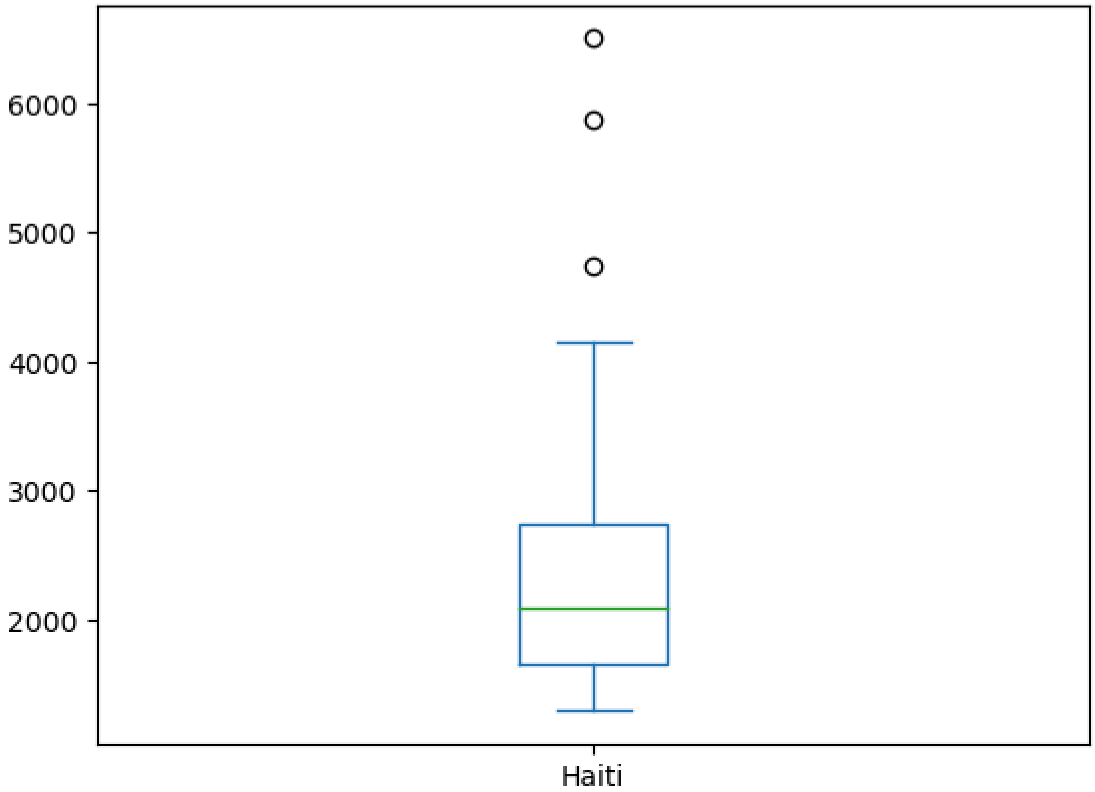
Pandas Function:

```
DataFrame.plot.box()
DataFrame.plot(kind = 'box')
```

Example:

```
df_can.plot(kind='box')
```

Visual:



Scatter Plot

Uses Cartesian coordinates to display values for two variables

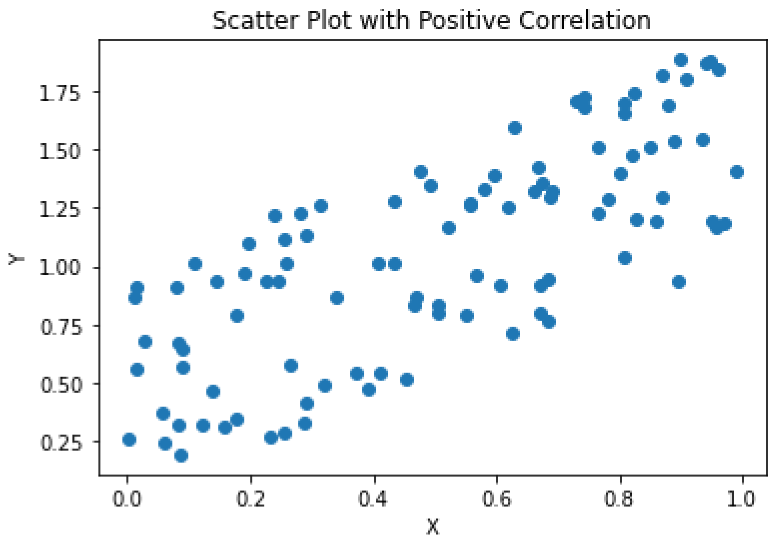
Pandas Function:

```
DataFrame.plot.scatter()
DataFrame.plot(x, y, kind = 'scatter')
```

Example:

```
df.plot(x='Height', y='Weight', kind='scatter')
```

Visual:



Plotting directly with Matplotlib

Line Plot

Shows trends and changes over time

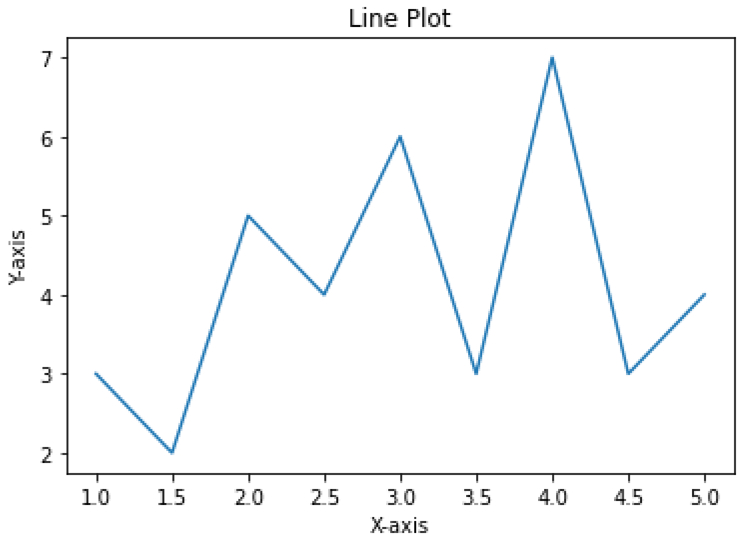
Matplotlib Function:

```
plt.plot()
```

Example:

```
plt.plot(x, y, color='red', linewidth=2)
```

Visual:



Area Plot

Display data series as filled areas

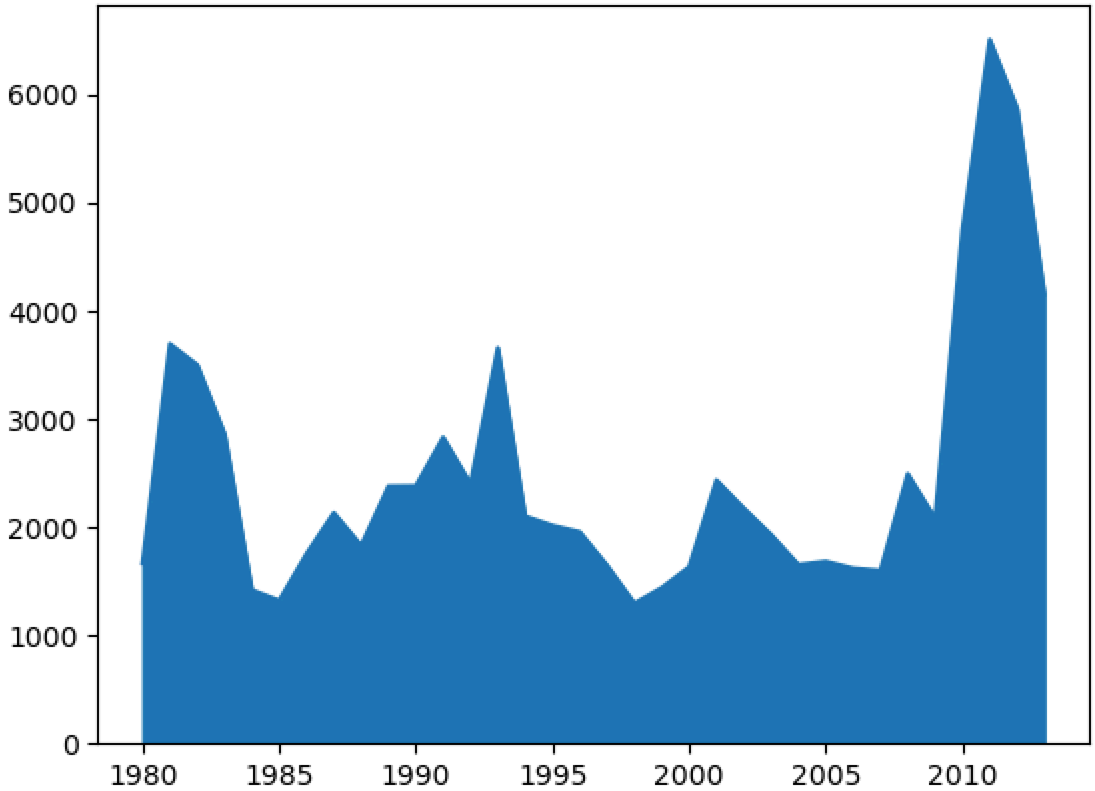
Matplotlib Function:

```
plt.fill_between()
```

Example:

```
plt.fill_between(x, y1, y2, color='blue', alpha=0.5)
```

Visual:



Histogram

Displays bars representing the data count in each interval/bin

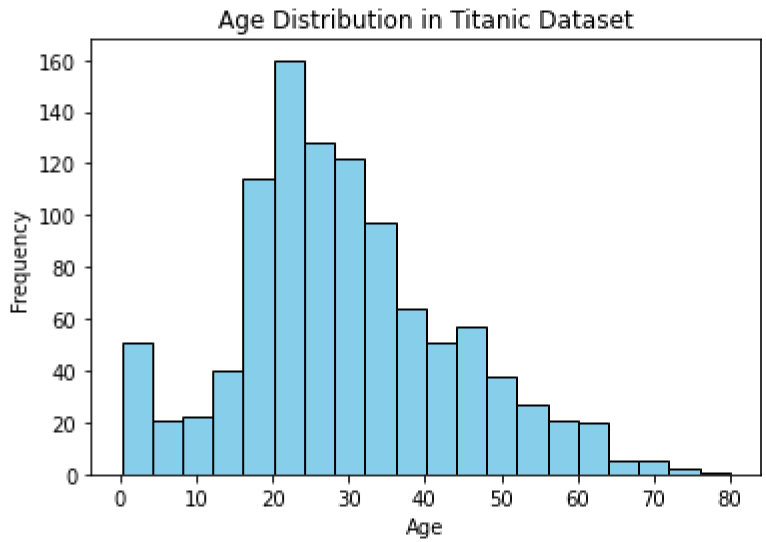
Matplotlib Function:

`plt.hist()`

Example:

`plt.hist(data, bins=10, color='orange', edgecolor='black')`

Visual:



Bar Chart

Displays data using rectangular bars

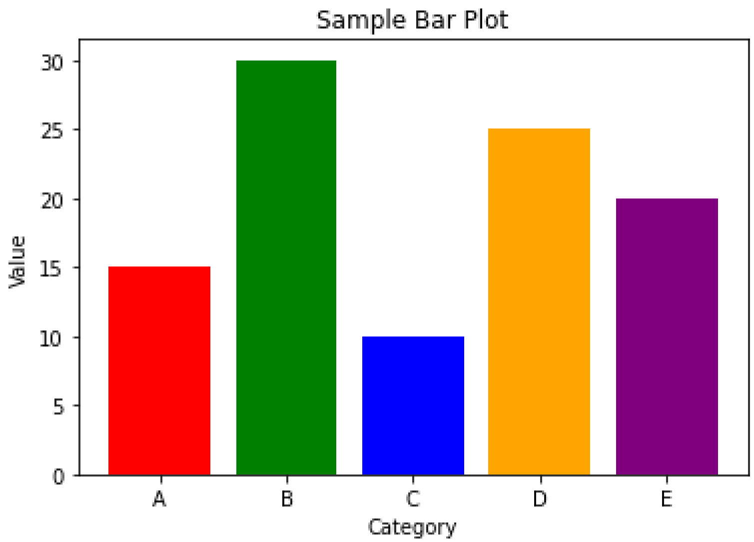
Matplotlib Function:

```
plt.bar()
```

Example:

```
plt.bar(x, height, color='green', width=0.5)
```

Visual:



Pie Chart

Displays data as a circular plot divided into slices, representing proportions or percentages of a whole

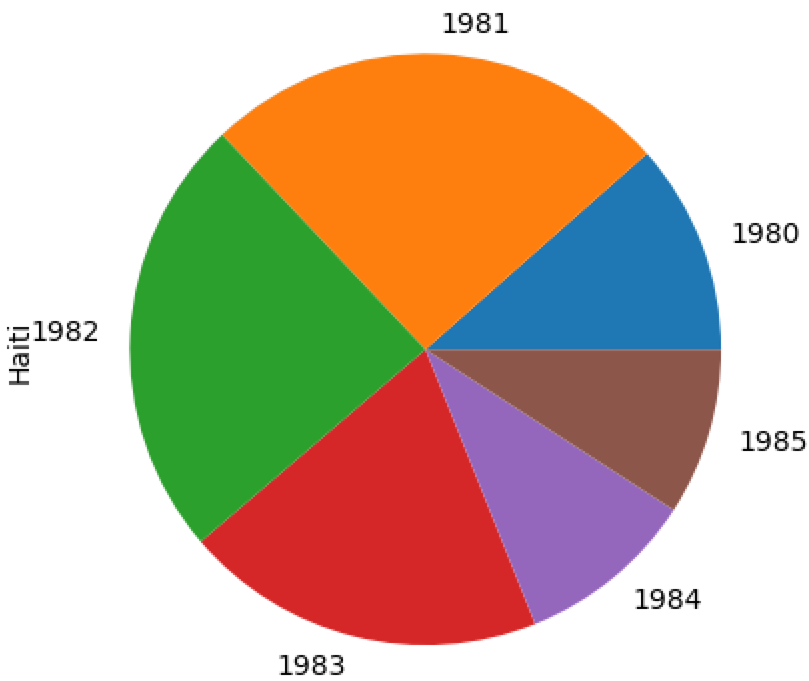
Matplotlib Function:

`plt.pie()`

Example:

```
plt.pie(sizes, labels=labels, colors=colors, explode=explode)
```

Visual:



Box Plot

Displays the distribution of a dataset along with key statistical measures

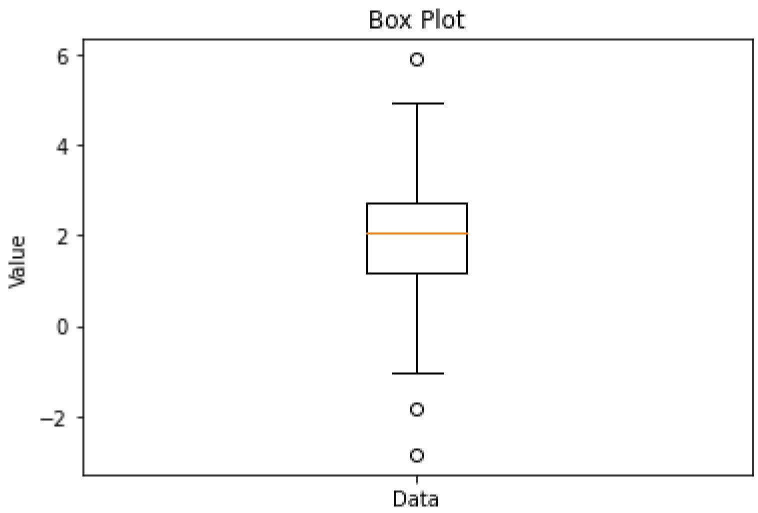
Matplotlib Function:

```
plt.boxplot()
```

Example:

```
plt.boxplot(data, notch=True)
```

Visual:



Scatter Plot

Uses Cartesian coordinates to display values for two variables

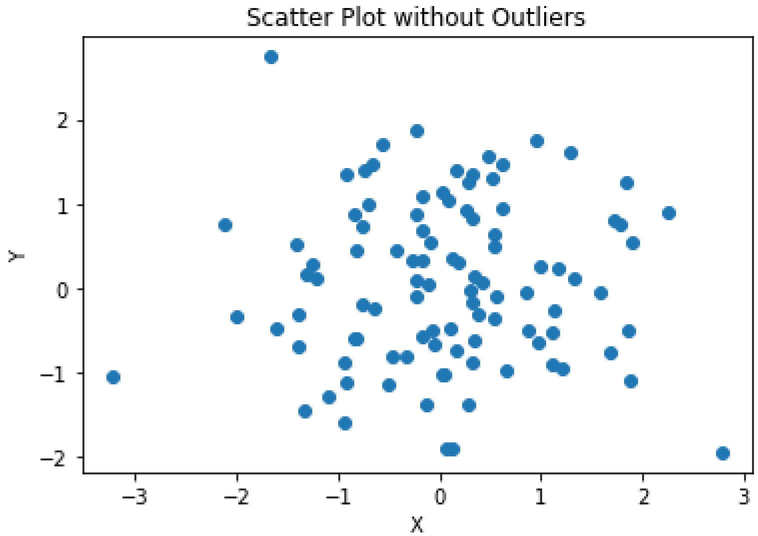
Matplotlib Function:

```
plt.scatter()
```

Example:

```
plt.scatter(x, y, color='purple', marker='o', s=50)
```

Visual:



Subplotting

Creating multiple plots on one figure

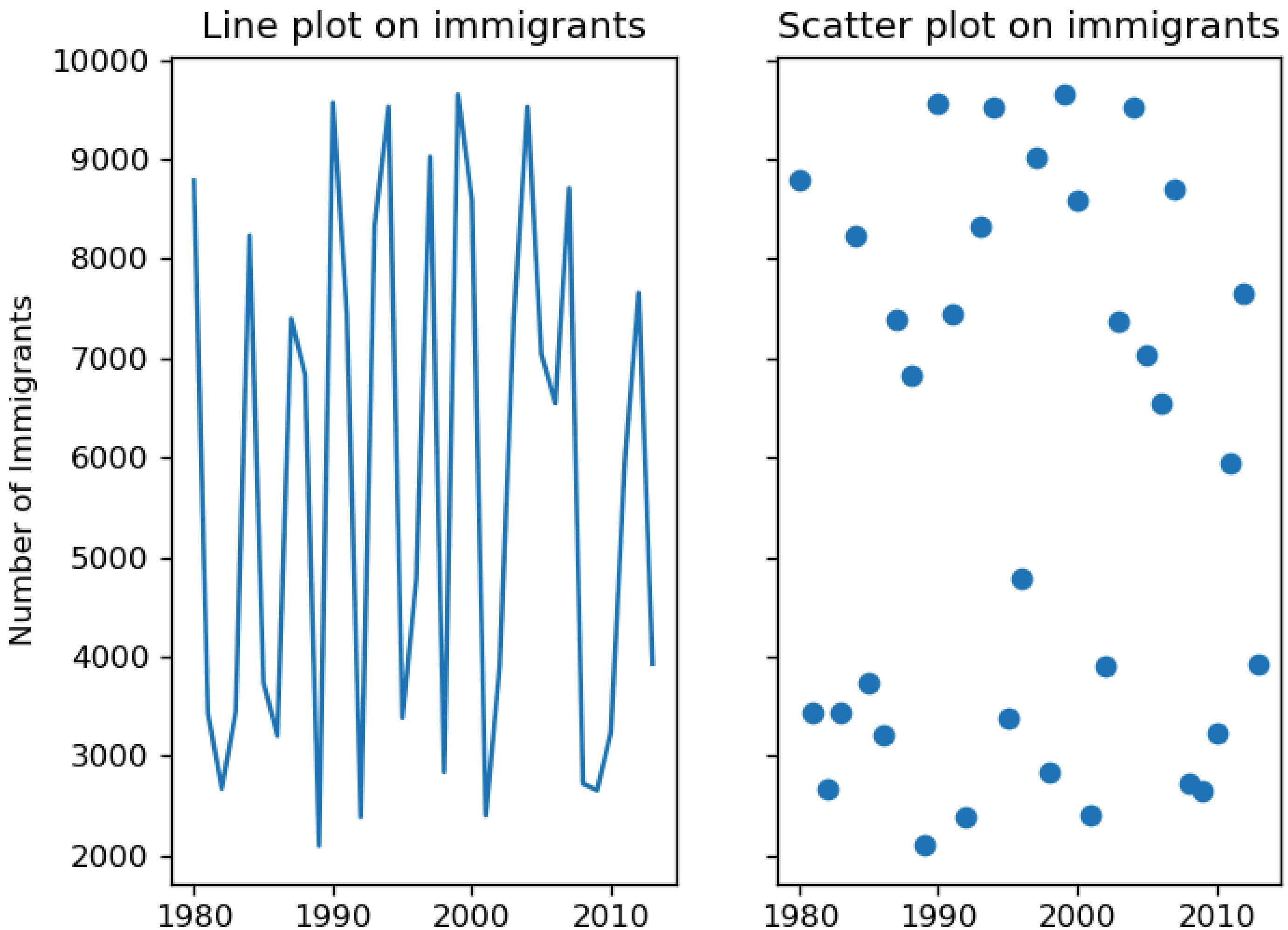
Matplotlib Function:

```
plt.subplots()
```

Example:

```
fig, axes = plt.subplots(nrows=2, ncols=2)
```

Visual:



Customization

Customizing plot, including adding labels, title, legend, grid

Matplotlib Function: Various customization

Example:

```
plt.title('Title')
plt.xlabel('X Label')
plt.ylabel('Y Label')
plt.legend()
plt.grid(True)
```

Visual: